

DiatomS mEEt Databases: resources and practices to enable large-scale ecological research (SEED)

Project report

ISDR type II ECR networking grant (2024-2025)

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Introduction

The International Society for Diatom Research (ISDR) granted its first ECR networking grant in 2023. The awarded project has the overall objective to improve the quality of open-access diatom data for enabling large-scale ecological synthesis, with special emphasis on historically under-presented geographical regions from Iberoamerica. To achieve this, the project's specific objectives were: 1) build trust within a community of data generators and stewards, 2) inspire the next generation of diatom ecologists to create, share, and utilize regional and continental scale diatom counts, and 3) stimulate networking among the Iberoamerican diatom researchers that will lead to future projects. This report serves to present the main achievements, results, finances, and lessons learnt of the grant.

Main achievements and synergies created

The SeeD project organized its first general meeting on September 25th, 2024, after launching the first call for participation and data contribution on February 2nd, 2024. A total of 35 participants from 6 different countries attended the meeting. The online meeting was recorded for those participants unable to attend and/or to give the possibility for revisiting the video when needed. The meeting was in Spanish due to its most common usage among the participants so far, yet presentation slides were in English and English was also agreed to be used if needed. The main result was the creation of the [SeeD inventory](#), a shared Google spreadsheet to collect metadata information from diatom records owned by the participants and willing to share, encompassing any type of aquatic environment and temporal scale. The objective was to build an initial inventory of published and unpublished, and if so, link this data with a DOI. Participants willing to share unpublished diatom datasets were the focus for the subsequent steps of the project, aiming at establishing a basic workflow to mobilize the data.

Another important result that came out from the first meeting was the establishment of the [Zenodo community folder of SeeD](#). We further discussed with the group the progress in two main aspects: 1) format and nomenclature of files and data to share, and 2) uploading unpublished data to Zenodo (repository that we will use to archive the different datasets that will

make up the first version of the database). We did encourage the use of Zenodo as a repository to archive datasets as constituents of the future SeeD database. It is not compulsory archiving datasets in Zenodo because they can already be published somewhere else. However, having a DOI is a prerequisite to include datasets in SeeD as we aim for open and fair data practices. In total, the SeeD Zenodo folder has 12 datasets, 9 restricted (visible but not publicly available), and 3 open. The workflow in archiving diatom datasets in Zenodo with a consistent data structure and metadata was documented and constituted one of the main results of SeeD before transitioning to the second year of the project.

Early 2025, we continued organising meetings with the coordinators to curate and maintain the SeeD inventory and Zenodo repository. We worked collaboratively with the individual data contributors to solve questions about uploading their datasets in Zenodo, formatting the excel templates, and discussing new types of datasets. For instance, several members of the network were interested in contributing data from paleolimnological records and marine sites. We also presented the SeeD project in three events: the diatom web academy (April 8th, 2025), the Young Diatomist online meeting during the South Africa IDS on July 8th, 2025, and during the introductory meeting of the 2nd call of the ISDR ECR grants. We were able to share lessons learned from our experience as part of the first cohort of funded projects.

In summer 2025, the coordinators contacted Dr [Manuel Chevalier](#), a freelance data scientist with an extended academic career working on several biological organisms for large-scale (paleo)ecological analyses. Dr Chevalier Manu's proven experience in organising online workshops and training events using the R software was also a plus for the project. We decided to spend the networking grant budget by hiring Dr Chevalier with a two-fold objective: 1) convert the SeeD dispersed datasets into a functional relational database, and 2) organize a 2-day workshop with the participants to visualize and analyse the database, as well as discuss the next steps.

In the months leading up to the final SeeD workshop (12-13 March 2026), the coordinators and Dr Chevalier met every other week to prepare the database, agreeing on the final number of tables, format, and structure (Fig 1). The number of datasets for the first version of the database was a moving target, and we decided to include both open and restricted datasets in Zenodo and a few extra additions such as diatom monitoring time series from Spanish river authorities. An important aspect of the workshop preparation phase was the nomenclatural harmonization of diatom names. From the initial number of 4776 unique diatom taxon names, we reduced the list to ca 2500 harmonized taxa by updating names to the most recent taxonomically accepted synonym (i.e., using the Omnidia database), curating consistency in taxon names, including misspellings, elevating undetermined taxa to spp, and maintaining low-level taxon uncertainties (cf., aff, type) to the original name. Importantly, the lookup table includes all the name changes in order to compare harmonised names with original names. Additional harmonization steps will strongly depend on the question(s) and trade-off between taxonomic loss (yet inevitable) and spp-environment relationships.

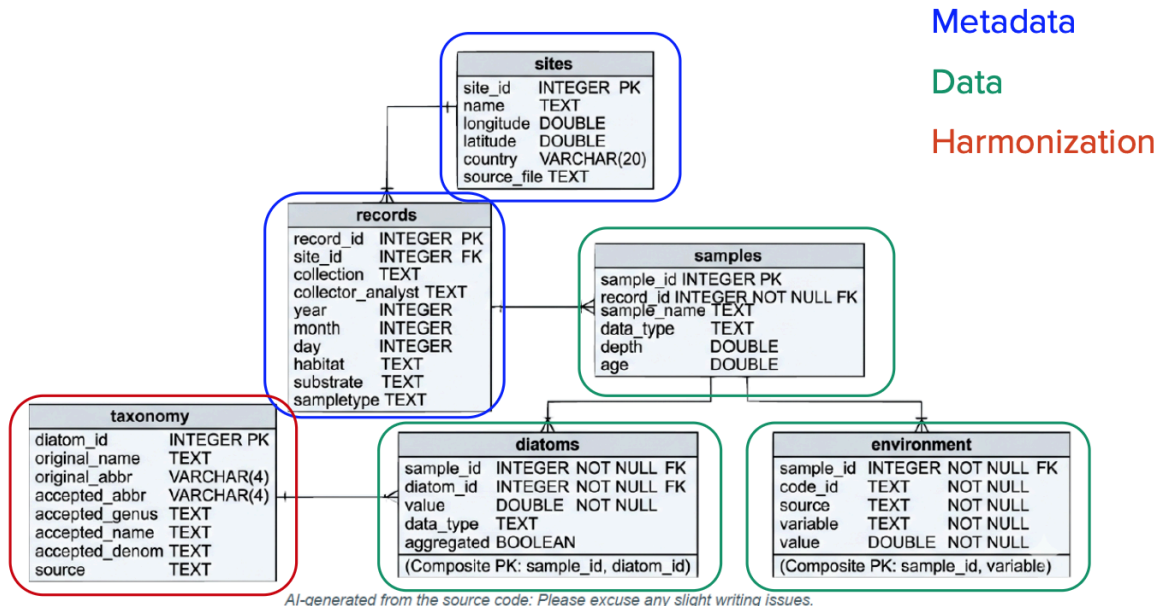


Fig 1. SeeD database structure showing the 6 constituent tables with associated descriptors

The final SeeD workshop was held online between 12 and 13 March. All data contributors were invited, with a total participation of 9-10 colleagues between the two workshop days (Fig 2); the workshop was recorded, and the language used was English. An invitation was also extended to Don Charles (Neotoma diatom steward) to fulfil another of the SeeD project objectives: create synergies among diatom networks elsewhere in Iberoamerica and learn from well established, community-curated database infrastructures.

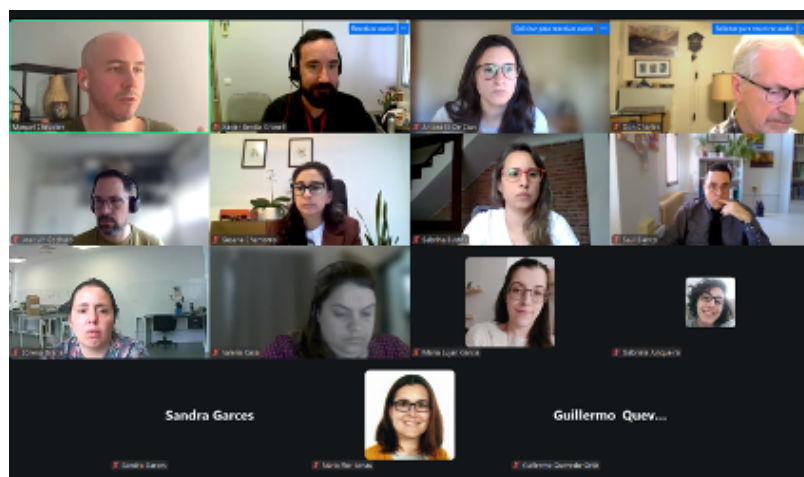


Fig 2. Proud participants of the SeeD workshop (12-13 March 2026).

During the workshop, the SeeD database v1 was presented (Fig 3). It includes: 1,478 sites, 1727 records (several sites have more than one record) from 8 countries, including 14 paleorecords. A total of 3449 diatom samples, with different data types: 2008 counts, 1378 percentages, 63 presence/absence, and 87620 diatom observations. A total of 8594 environmental observations form the basis of basic water chemistry data.

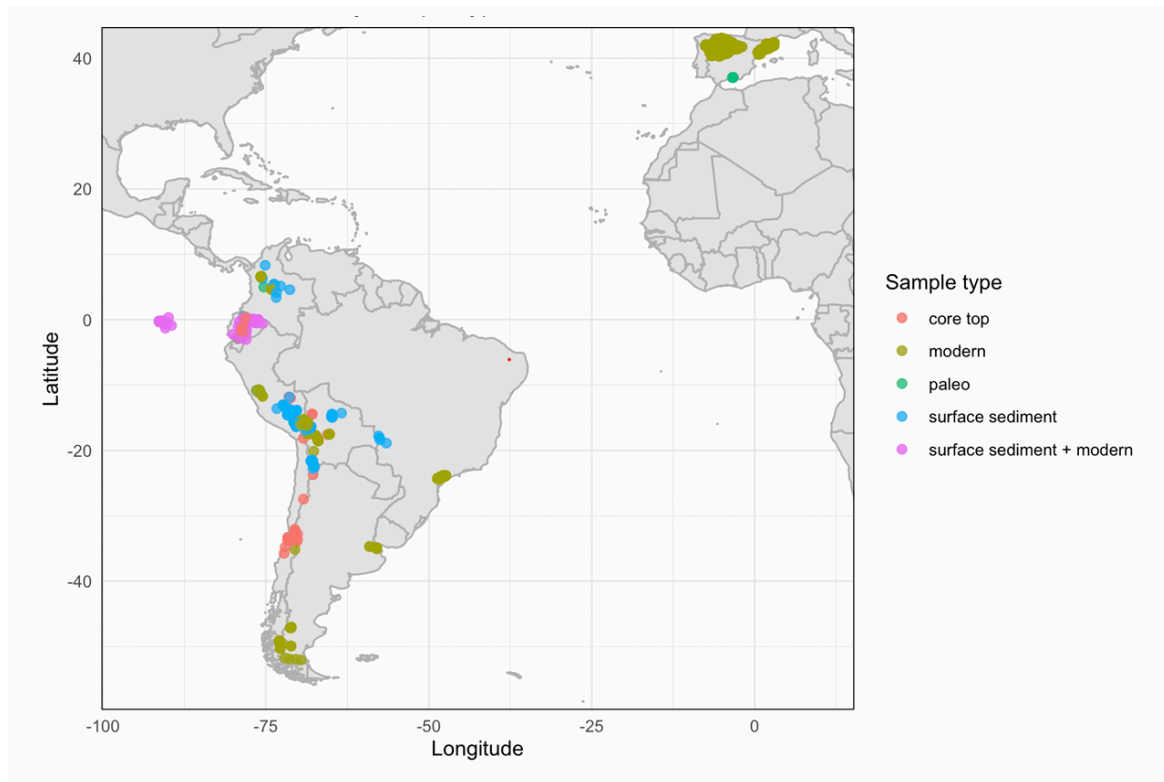


Fig 3. Distribution of the diatom sites coloured by sample type in the SeeD database.

The workshop included two breakout sessions about a) elaborating on collaborative projects that may interrogate the SeeD database for (paleo)ecological studies, and b) curating the diatom master taxon table. Dr Chevalier showed the R functions specifically created for SeeD for getting data out of the database and obtaining diatom datasets in a usable format in R. The workshop ended with a group discussion to delineate the next steps for the network: SeeD what is missing in the database (e.g., metadata about publications), what would be feasible and relevant to analyze, and initiatives for a SeeD v2 (i.e., new dataset contributions).

Lessons learnt and future activities

The SeeD project proved successful in mobilizing diatomists working in and/or from Ibero-America who have long been interested in sharing their data but lacked training in data formatting, management, and analyses. SeeD not only established a network from early-career to more advanced researchers to foster new collaborations in diatom (paleo)ecology, but also provided tools for better curation of large-scale diatom datasets—for instance, a diatom taxon master list that can be complemented with teaching courses on data stewardship. We learned how challenging it can be to resolve taxonomic inconsistencies without creating endless

discussions over species identification if a voucher flora is not available. Equally important, we also learnt that the diatom convoluted taxonomy should not stop synthesis projects like SeeD. When scientific results are not the primary objective of such networking grants, other outputs, perhaps less tangible, may be more impactful than a scientific paper.

Future activities should focus on maintaining and enlarging the diatom taxon list by adding more synonyms and validating diatom names using a combination of community-based curation methods and reproducible workflows in R. SeeD is currently joining efforts with the Spanish-speaking network of diatomists (ReDiH) aiming to organize small courses on data stewardship. An early proof of this engagement is the organization of the workshop “Diatom ecology through lenses of open and FAIR practices” at the 16th European Diatom Meeting, Murcia, Spain (4–9 July 2026). Workshop materials and R scripts are available on [Github](#). One of the coordinators (XB) also presented the SeeD project at the EDM and highlighted future integration plans with Neotoma, one of the most prominent community-curated paleoecology databases. It is foreseen that SeeD, as part of the Tropical South American Diatom Database constituent database of Neotoma, will be integrated into the database, ensuring that most, if not all, diatom datasets will be archived and therefore available to the global diatom community. Therefore connecting Iberoamerican diatom datasets and stewards with North America and Europe.

Looking ahead, SeeD will be presented at the 28th International Diatom Symposium (9-13 August 2027, Prague Czechia). Potential results might include: understanding the variation of unclassified diatom taxa in the database, a best practices document to mobilize, share, and analyze diatom datasets, biogeographic distributions of ecologically important diatom taxa, and general relationships between diatom abundance and aquatic habitat types. We oversee two types of publications: one more technical, describing the database and its potential applications (to be submitted to Diatom Research by the end of 2027), and one detailing the general workflow in data sharing using open and fair practices and lessons learned during the project.